

Certified analytical bounds for hypersonic reachable sets

Ethan Knox

Independent Researcher, Champaign, IL
ethank5149@gmail.com

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Abstract

This is a proposal in nonlinear analysis and dynamical systems. The object is a singularly perturbed family of differential inclusions in which ordinary time-parametrised limits do not retain the concentrated state transitions: as the parameter vanishes, the motion carrying the important transitions occupies a vanishing time, so it disappears from those limits while still deciding where the trajectories go. Compactness itself does not fail — the corridor is compact and the occupation measures stay weak-* compact. What fails is the ability of a time-parametrised limit to keep a transition of vanishing duration, since it weighs an interval of time by its length. The natural structural language for this is a *profile decomposition* — a low-dimensional *skeleton*, finitely many concentrating *bubbles*, a *residual* bounded in a stated metric — and the question is whether that is a theorem about solution sets or only a description of them. It is deliberately not called a remedy: the theorem below needs none of that machinery, which is what makes its necessity a question.

We state it as a hierarchy. *Theorem A*: every sequence of trajectories splits, along a subsequence, into a skeleton, finitely many rescaled profiles above any fixed size, and a remainder. *Theorem B*: the profiles induce a set-valued jump map. *Theorem C*: the solution set of the family lies in that of the skeleton-and-jump system, thickened by an explicit ball. This document proves Theorem C, with every constant written down, for a model whose potential is asymmetric in the one way that produces concentration.

It does not prove Theorem A, and the relation wants stating exactly: the theorem proved here establishes no decomposition and its proof uses none. What it does is twofold. It is the quantitative estimate any decomposition must reproduce — including the profile count and the amplification they compound, which a Lyapunov functional supplies *a priori* where the classical argument obtains them by contradiction and without a constant. And it shows the concentration structure can be reached without the decomposition machinery at all, by an exact integral identity and a budget. Whether that machinery buys anything the elementary route does not is the first question of the programme.

Hypersonic entry supplies the problem and is the test the result is measured against; it is not the subject. Only the theorem of [Section 4](#) is claimed as proved.

1 The question, reduced

The object we want enclosed is the solution set of a family of differential inclusions: every state reachable under every admissible control and every admissible selection. Two things make that hard. Its dimension puts the exact methods out of reach; and the bound must come with a proof, which is to be a pair of functions whose superlevel set contains the solution set, exhibited and checked rather than produced by a solver ([Section 6](#)). This document is narrower than the programme and prior to it: it takes the analytical difficulty the programme turns on, strips it of the application, and proves a theorem about it.

The difficulty is a singular limit that defeats the ordinary compactness arguments. For atmospheric entry the parameter is $\varepsilon = H_s/R_e$, the ratio of scale height to planetary radius, whose square root is the ratio of the oscillation period about equilibrium to the slow time of the flight. As $\varepsilon \rightarrow 0$ the motion separates into arcs where the forcing is not felt and *passes* where it is felt strongly for a vanishing time. A pass becomes instantaneous and keeps its effect, so in the limit a trajectory is a sequence of arcs joined by jumps — yet the natural limiting objects do not see the jumps, since they weigh an interval by its length and a pass has none. A solution set is a support, not an integral, so what has no mass still counts.

The language for this is a profile decomposition, obtained in the theory of nonlinear dispersive equations by iterating the dichotomy of the concentration-compactness principle [\[1, 2\]](#). It delivers structure — which pieces

can occur, and that nothing else can, by contradiction — and no constant: the authors of the wave-equation theory are explicit that the uniform bound following from their decomposition carries a constant they cannot describe [2, Corollary 2]. Our proposal is that here the structure can be had *quantitatively*, because the family carries a Lyapunov functional of bounded total variation, uniformly in the parameter. That bounds in advance the number of profiles above any fixed size, the time the motion may spend concentrating, and the accumulated growth of the comparison. The count is Eq. (3.15), proved below; the decomposition it would organise is not.

Remark 1.1 (What the small parameter is, and what it is not). It is tempting to write the reduced problem as a slow state driven by a fast one, $\dot{z} = g(z) + h(y)$ with $\varepsilon \dot{y} = q(y, z)$ and h bounded. That form does not contain the difficulty: an excursion in y of duration $O(\varepsilon)$ under a forcing of order one contributes $O(\varepsilon)$ to z , leaving no trace in the limit, and ordinary averaging describes everything. What produces a jump is an *asymmetric potential* together with a scaling that keeps the oscillation slow while the asymmetry sharpens: the air supplies a wall steepening without bound below the equilibrium while gravity is uniform above it, and because ε enters only as the *width of the band* in which the air is felt, a rebound occupies a vanishing fraction of its oscillation while delivering an impulse of order one. Lemmas 3.3 and 3.4 are those two statements, and each says what measuring z in band thicknesses instead would have given. Concentration is the asymmetry of the well read at the right scale, not an assumption about the size of a force.

1.1 What this document does not prove

Theorem 4.2 is Theorem C, for one model. Three statements belong to the programme and are not proved here, and each is stated in full in Section 8. *Theorem A*, the decomposition itself, of which Theorem 4.2 asserts nothing: what Assumption 2.4 supplies is the count Eq. (3.15), and proving the decomposition it would organise is *objective 1*. *The limit of the family in the weak topology*, defined in Section 6 and owed by *objective 4*. And *the passage from the weak limit back to the support*, the hardest link in the chain: granting the limit of the measures and the profiles it carries, that the limiting solution set lies in the one those profiles generate is a further implication, which the proof below avoids by getting its inclusion from trajectories directly. Which route the general case should take is open, and it is the question we would most value an advisor's judgment on. A fourth obligation is deferred rather than open: a bound in closed form must finally be *evaluated*, and a guarantee does not ordinarily survive floating point, the setting in which it does being validated interval arithmetic [3, Ch. 3]. That concerns evaluating a bound rather than proving one, and we claim no computation.

1.2 Terminology

Two fields meet here and several words mean different things in each; we fix them once.

Orbit. One closed curve of Eqs. (2.1) to (2.2) at *frozen* (c, θ) and zero damping (Lemma 2.8). Not a trajectory about the planet, and not a piece of an actual trajectory, which closes no orbit: orbits serve only as the comparison objects defining P_0, D_0 in Lemma 3.3. An actual trajectory is decomposed by Definition 3.5, which cuts it at its crossings of $z = 0$ into *passes* ($z \leq 0$) and *arcs* ($z \geq 0$). **Selection θ .** One member of the admissible family of fields, *fixed along a trajectory* and quantified over at the end (Remark 2.3); never a function of time in what is proved here. **Profile m_θ versus density m_θ^ε .** The profile is the ε -free function of Assumption 2.1, read in units of the band's thickness; the density is the profile as the motion meets it, $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon)$. Hypotheses are placed on the profile, which is why they are uniform in ε . **Lyapunov functional.** Two objects wear the name and do opposite jobs. The dissipation Θ_d of Assumption 2.4 is non-increasing along the motion and measures what it spends; the amplitude W is *not* monotone, the control being able to pump it, but obeys the *a priori* bound Eq. (3.2). Every count below rests on that pair. The *dual pair* (Φ, Ψ) of Section 6 is non-decreasing, and $\{\Psi \geq 1\}$ encloses the reachable set; that pair is the certificate, and exhibiting one is a proof whether or not it is ever evaluated. The first kind is in hand; constructing the second is *objective 4*. **Averaging.** Always of the fast part, with a single phase (Lemma 4.6); over *several* phases, whose frequencies may cross, it is a different problem, attempted nowhere below, and is *objective 3*. **The oscillator.** Always the motion of Eqs. (2.1) to (2.2), the altitude about the glide; the *attitude* oscillator of *objective 3*, three orders of magnitude faster, is named in full.

2 The model and its hypotheses

Fix an integer $d \geq 1$ and a horizon $T > 0$. The state is $x = (z, v, s)$, with $z \in \mathbb{R}$ an altitude, $v \in \mathbb{R}$ its rate, and $s \in \mathbb{R}^d$ a vector of slow coordinates. The system is

$$\dot{z} = v, \tag{2.1}$$

$$\dot{v} = c m_\theta^\varepsilon(z) - 1 - \kappa(s, c, \theta) v, \tag{2.2}$$

$$\dot{s} = F(s, c, \theta) + m_\theta^\varepsilon(z) \Gamma(s, c, \theta). \quad (2.3)$$

Here $c(\cdot)$ is the control and $\theta \in \Theta$ is a *selection* — a choice of admissible field from a bounded family, fixed along each trajectory and quantified over at the end (Remark 2.3). The small parameter $\varepsilon \in (0, 1]$ appears in one place only, inside the density: the *profile* m_θ is read on the layer's own scale,

$$m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon), \quad M_\theta^\varepsilon(z) = \varepsilon M_\theta(z/\varepsilon), \quad (2.4)$$

where M_θ is the antiderivative of m_θ with $M_\theta(0) = 0$, so that $(M_\theta^\varepsilon)' = m_\theta^\varepsilon$ and $M_\theta^\varepsilon(0) = 0$. The *potential* is

$$\Phi_{c,\theta}(z) = z - c M_\theta^\varepsilon(z), \quad \text{so} \quad \Phi'_{c,\theta}(z) = 1 - c m_\theta^\varepsilon(z), \quad (2.5)$$

and Eqs. (2.1) to (2.2) are a unit-mass particle in that potential with a weak damping. We write $\Phi_{c,\theta}$ without decoration, though it depends on ε : all of that dependence is the density's, and Lemma 3.3 is where it is unpacked. The *amplitude* is its energy,

$$W = \frac{1}{2}v^2 + \Phi_{c,\theta}(z). \quad (2.6)$$

The whole of Section 3 is an account of one object: the shape of this potential.

That ε sits inside the density and nowhere else is the whole of the scaling, and Remark 1.1 says why it is the choice that produces concentration.

Assumption 2.1 (H1: the wall). For each $\theta \in \Theta$ the profile $m_\theta \in C^1(\mathbb{R})$ satisfies $m_\theta > 0$ and $m'_\theta < 0$; $m_\theta(\xi) \rightarrow 0$ as $\xi \rightarrow +\infty$ with $\int_0^\infty m_\theta < \infty$; $m_\theta(\xi) \rightarrow \infty$ as $\xi \rightarrow -\infty$; and $1/c$ lies in the range of m_θ for every $c \in [c_{\min}, c_{\max}]$. We require in addition that these hold *uniformly* over Θ : $m_+ := \sup_\Theta m_\theta(0) < \infty$ and $M_\infty := \sup_\Theta \int_0^\infty m_\theta < \infty$. Compactness of Θ does not by itself give either, and both are used as finite constants throughout, so they are asked for here rather than inferred.

The two one-sided conditions are the whole physical content: the air thins upward without bound on the integral, and thickens downward without limit. Nothing in what follows uses the shape of m_θ beyond Assumption 2.1, which is why θ may select it: an enclosure of an uncertain density is exactly a bounded family of admissible profiles. Because m_θ decreases, m_+ bounds m_θ^ε at and above the section $z = 0$ for every ε , and that is the only place a pointwise bound on the density is used below; inside the band m_θ^ε is unbounded as $\varepsilon \rightarrow 0$ and none is taken, what is used there being its *impulse*, which Assumption 2.4 bounds.

Assumption 2.2 (H2: data). $[c_{\min}, c_{\max}] \subset (0, \infty)$, Θ and $S \subset \mathbb{R}^d$ are compact. The maps $F, \Gamma: S \times [c_{\min}, c_{\max}] \times \Theta \rightarrow \mathbb{R}^d$ and κ are continuous, bounded by $F_{\max}$, $\Gamma_{\max}$ and $\kappa_{\max}$, Lipschitz in s uniformly in (c, θ) with constants κ_F, κ_Γ , and Lipschitz in c uniformly in (s, θ) with constant L_c . The last is needed because c varies in time and Lemma 4.6 differentiates Γ along the motion; continuity in c alone would not support it. Moreover $\kappa > 0$. The selection θ is a fixed element of Θ along each trajectory; only $c(\cdot)$ varies with time.

Remark 2.3 (Why the selection is fixed, and what that costs). The reachable set below is the union over $\theta \in \Theta$ of the sets reached at fixed θ : the uncertainty is quantified over, not allowed to vary *during* a trajectory. The reason wants stating, since it is where a hidden hypothesis would otherwise sit. The potential Eq. (2.5) depends on θ , so a time-varying θ would move the well beneath the amplitude Eq. (2.6) and Eq. (3.1) would acquire a term in $\partial_\theta M_\theta^\varepsilon(z) \dot{\theta}$ — which for a merely measurable $\theta(\cdot)$ one cannot bound, or in general even write. Every estimate descending from the amplitude would go with it.

The budget does not depend on this. Assumption 2.4 asks for one functional decreasing for *all* (s, c, θ) at once, so Eq. (5.1) holds for an arbitrary measurable $\theta(\cdot)$ as it stands. The division is clean: the slow budget survives a time-varying selection, the fast amplitude law does not.

The cost is real and runs the unsafe way. The set reached when the atmosphere and the coefficients may vary in time is *larger* than the union taken here, so Theorem 4.2 bounds the union and not that larger set. Enlarging the class to selections of bounded variation in altitude — what the physics supplies, the atmosphere being smooth in space and the mass lost monotonically — is objective 2, an enlargement of the enclosure rather than a refinement of its constants.

Assumption 2.4 (H3: a Lyapunov functional). There is $\Theta_d \in C^1(\mathbb{R}^d)$ and a constant $c_0 > 0$ with

$$\nabla \Theta_d(s) \cdot F(s, c, \theta) \leq 0, \quad \nabla \Theta_d(s) \cdot \Gamma(s, c, \theta) \leq -c_0,$$

for all (s, c, θ) . Write $M = (\max_S \Theta_d - \min_S \Theta_d)/c_0$.

A Lyapunov functional for the slow motion, non-increasing along every trajectory for every control and every selection. Bounding its total variation over the compact corridor converts monotonicity into a count: by Eq. (5.1) it bounds $\int m_\theta^\varepsilon d\tau$ uniformly in ε , and every count in Section 5 descends from that one inequality. In the application it is the specific mechanical energy, falling at the drag power because neither the lift nor the Coriolis force does work.

Assumption 2.5 (H4: control). $c(\cdot)$ is absolutely continuous with values in $[c_{\min}, c_{\max}]$ and $|\dot{c}| \leq \dot{c}_{\max}$. We also name the sharper form a finite guidance interval supplies:

$$(H4') \quad c \text{ is piecewise constant on intervals of length } \geq \Delta, \quad (2.7)$$

with jumps $|\Delta c| \leq |\Delta c|_{\max}$ at the at most T/Δ switching times, so that $\dot{c} = 0$ between switches and Eq. (3.1) reads $\dot{W} = -\kappa v^2$ there. The difference between the two is not cosmetic: by Remark 4.4 it decides whether the residual vanishes with ε at all.

Assumption 2.6 (H5: corridor). Every trajectory remains on $[0, T]$ in a compact corridor $\mathcal{X} = \{|z| \leq z_{\max}, |v| \leq v_{\max}, s \in S\}$, and $x(0) \in \mathcal{X}_0 \subseteq \mathcal{X} \cap \{z \geq 0\}$: the motion begins at or above the section, where the antiderivative is bounded by εM_{∞} , so that

$$|W(0)| \leq W_{\circ} := \frac{1}{2}v_{\max}^2 + z_{\max} + c_{\max}M_{\infty}. \quad (2.8)$$

The corridor is a fixed box, which the scaling of Eq. (2.4) is what permits: the oscillations of interest have amplitude of order one, so bounding them costs nothing that shrinks. Note what the box does *not* give. The amplitude Eq. (2.6) is not bounded by $\frac{1}{2}v_{\max}^2 + z_{\max}$, because $M_{\theta}^{\varepsilon}(z) < 0$ for $z < 0$ makes $-cM_{\theta}^{\varepsilon}(z)$ positive there and the antiderivative is unbounded below as $\varepsilon \rightarrow 0$ at fixed $z < 0$. The ceiling on the amplitude is therefore not read off the corridor at all; it is derived in Lemma 3.1 from the amplitude law itself, and the bound on M_{θ}^{ε} follows from it in Eq. (3.3) rather than the other way about.

Remark 2.7 (What is uniform, and what the hypotheses are not). Every constant named above is a bound over $S \times [c_{\min}, c_{\max}] \times \Theta$. No constant below depends on the particular control, the particular selection, or the particular initial point: that is the sense in which the result is uniform over the admissible family, and it is why an enclosure of an uncertain field is the natural setting rather than a complication. Note also what is absent. There is no transversality condition, no non-grazing condition, and no assumption that a pass is a single dip — the last is Lemma 2.8, a consequence of $m'_{\theta} < 0$. Every hypothesis is a condition on the data, checkable before any trajectory is computed.

Lemma 2.8 (One well, and one dip per period). *Under Assumption 2.1 the potential $\Phi_{c,\theta}$ is strictly convex, with*

$$\Phi''_{c,\theta}(z) = -\frac{c}{\varepsilon} m'_{\theta}(z/\varepsilon) > 0, \quad (2.9)$$

and attains its minimum at the unique $z_(c, \theta) = \varepsilon \xi_*$, where ξ_* is the unique point with $cm_{\theta}(\xi_*) = 1$: the glide sits inside the band, at height of order ε . Consequently every sublevel set $\{\Phi_{c,\theta} \leq W\}$ is a bounded interval $[z_-(W), z_+(W)]$, and at frozen (c, θ) every bounded motion of Eqs. (2.1) to (2.2) with $\kappa = 0$ is a single closed orbit — one closed curve in the (z, v) plane, traversed periodically — on which z attains exactly one minimum and one maximum.*

Proof. Differentiating Eq. (2.5) twice with Eq. (2.4) gives Eq. (2.9), positive because $c, \varepsilon > 0$ and $m'_{\theta} < 0$. A strictly convex C^2 function has at most one stationary point, and $\Phi'_{c,\theta} = 1 - cm_{\theta}^{\varepsilon}$ changes sign exactly once since m_{θ} decreases strictly from $+\infty$ to 0 with $1/c$ in its range; so z_* exists and is unique. Convexity makes each sublevel set an interval, bounded because $\Phi_{c,\theta} \rightarrow +\infty$ at both ends. At frozen (c, θ) and $\kappa = 0$, Eq. (2.6) is constant, so the orbit is the level curve $\{\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)\}$ — a single closed curve meeting $v = 0$ exactly at the interval's endpoints, which are the one minimum and one maximum of z . $\square$

Lemma 2.8 is worth noting for what it is not: a hypothesis. That a pass is a single dip rather than a train of them is a property of the motion, and here it costs one sign on the density rather than a condition imposed on solutions.

3 Three regimes of one oscillator

Glide, phugoid and skip are not three systems but three energies in the single well of Lemma 2.8. This section says which is which. The statement that organises everything is that ε sets no shape, only a clock — and that the clock is read at amplitude W/ε , which is what makes a pass concentrate rather than average away.

Lemma 3.1 (The amplitude law). *Along every trajectory of Eqs. (2.1) to (2.3),*

$$\dot{W} = -\kappa(s, c, \theta) v^2 - M_{\theta}^{\varepsilon}(z) \dot{c}, \quad (3.1)$$

whatever the profile m_{θ} and whatever ε ; the only time-dependent datum the potential carries is c , which is why Eq. (3.1) closes (Remark 2.3). Consequently, with

$$B = \frac{\dot{c}_{\max}}{c_{\min}}, \quad A = \kappa_{\max} v_{\max}^2 + B(z_{\max} + \frac{1}{2}v_{\max}^2),$$

the amplitude obeys the a priori bound

$$|W(\tau)| \leq W_{\circ} e^{B\tau} + \frac{A}{B} (e^{B\tau} - 1) \leq W_{\max} \quad \text{for } \tau \in [0, T], \quad (3.2)$$

where $W_{\max}$ is the right side at $\tau = T$ and $W_{\circ}$ is Eq. (2.8). (At $\dot{c}_{\max} = 0$ the middle expression is read as its limit $W_{\circ}$, and W is then non-increasing.) Consequently

$$M_{*} := M_{\infty} + \frac{z_{\max} + \frac{1}{2}v_{\max}^2 + W_{\max}}{c_{\min}} \quad (3.3)$$

bounds $|M_{\theta}^{\varepsilon}|$ wherever a trajectory can reach, uniformly in ε .

Proof. Differentiate Eq. (2.6) along Eqs. (2.1) to (2.2):

$$\dot{W} = v\dot{v} + \Phi'_{c,\theta}(z)\dot{z} + \frac{\partial\Phi_{c,\theta}}{\partial c}\dot{c} = v(c m_{\theta}^{\varepsilon} - 1) - \kappa v^2 + (1 - c m_{\theta}^{\varepsilon})v - M_{\theta}^{\varepsilon}(z)\dot{c},$$

where $\partial\Phi_{c,\theta}/\partial c = -M_{\theta}^{\varepsilon}(z)$ by Eq. (2.5). The two terms carrying $c m_{\theta}^{\varepsilon}$ cancel identically, which is Eq. (3.1); the cancellation uses only that the same $c m_{\theta}^{\varepsilon}$ appears in Eq. (2.2) and in $\Phi'_{c,\theta}$. For Eq. (3.2), eliminate the antiderivative in favour of the amplitude itself, which is what keeps the argument free of any prior ceiling on W : by Eqs. (2.5) and (2.6), $M_{\theta}^{\varepsilon}(z) = (z - \Phi_{c,\theta}(z))/c = (z - W + \frac{1}{2}v^2)/c$, so on the corridor $|M_{\theta}^{\varepsilon}(z)| \leq (z_{\max} + \frac{1}{2}v_{\max}^2 + |W|)/c_{\min}$. Feeding this into Eq. (3.1) and using Assumption 2.5, $|\dot{W}| \leq \kappa_{\max}v_{\max}^2 + B(z_{\max} + \frac{1}{2}v_{\max}^2) + B|W| = A + B|W|$, a closed linear differential inequality in $|W|$ alone. Gronwall and Eq. (2.8) give Eq. (3.2), and Eq. (3.3) is the displayed bound on $|M_{\theta}^{\varepsilon}|$ evaluated at $|W| \leq W_{\max}$. $\square$

It is worth being exact about what W is. It is *not* a Lyapunov functional and it is not monotone: by Eq. (3.1) the control pumps it whenever $M_{\theta}^{\varepsilon}(z)\dot{c} < 0$. What it is, is a controlled energy with an *a priori* growth bound, Eq. (3.2), which Assumption 2.5 makes a statement about the data rather than a hypothesis about solutions. The genuinely monotone quantity is the dissipation of Assumption 2.4. The counts below rest on the pair: one monotone, one merely bounded.

3.1 The mean density, and the time the air is felt

Recall from Section 1.2 that an *orbit* here is a closed curve of the *frozen, undamped* system, never a piece of an actual trajectory. Two exact identities do much of the quantitative work, and they have different standings: the first holds on such an orbit and comes from periodicity, the second holds along the trajectory as it is and comes from the budget alone.

Lemma 3.2 (The mean density of an orbit). *Let $\langle \cdot \rangle$ denote the time average over one closed orbit of Eqs. (2.1) to (2.2) at frozen (c, θ) and $\kappa = 0$. Then*

$$\langle m_{\theta}^{\varepsilon} \rangle = \frac{1}{c}, \quad (3.4)$$

for every amplitude W , every profile satisfying Assumption 2.1 and every ε ; the impulse of such an orbit equals its period divided by c . Separately, and along the trajectory as it is rather than along any frozen orbit, Eq. (5.1) bounds the time the motion may spend where the air is appreciable: for every $\mu > 0$,

$$|\{\tau \in [0, T] : m_{\theta}^{\varepsilon}(z(\tau)) \geq \mu\}| \leq \frac{M}{\mu}. \quad (3.5)$$

Proof. Integrate Eq. (2.2) over one period at $\kappa = 0$. The left side is $[v] = 0$ because v is periodic, so $\int (c m_{\theta}^{\varepsilon} - 1) d\tau = 0$, that is $c \int m_{\theta}^{\varepsilon} d\tau = \int d\tau$, which is Eq. (3.4) on dividing by the period. The impulse of the orbit is $\int m_{\theta}^{\varepsilon} d\tau = \langle m_{\theta}^{\varepsilon} \rangle \times (\text{period}) = (\text{period})/c$. For Eq. (3.5), the set where $m_{\theta}^{\varepsilon} \geq \mu$ contributes at least μ times its measure to $\int_0^T m_{\theta}^{\varepsilon} d\tau$, which Eq. (5.1) bounds by M ; this uses no periodicity and holds for an arbitrary measurable selection. $\square$

Equation (3.4) is the reason the skeleton is the glide: whatever the oscillation does, its time-averaged density equals the density at the equilibrium, so the glide is not merely the bottom of the well but the *mean* of every oscillation about it. Since $\langle m_{\theta}^{\varepsilon} \rangle$ does not depend on W , neither does the averaged slow motion, and the amplitude is an output of the reduced description rather than a driver of it. Note the standing of the identity: it concerns the frozen, undamped system, and the proof nowhere assumes the trajectory follows such an orbit. It supplies the *form* of the averaged field in Definition 4.1, and Lemma 4.6 estimates the error of using that field without any appeal to periodicity.

3.2 What ε does, and what it does not

Lemma 3.3 (ε is a clock, and the amplitude it is read against). *Let $\xi = z/\varepsilon$ and $u = v/\sqrt{\varepsilon}$, and let $\Phi_{c,\theta}^0(\xi) = \xi - cM_\theta(\xi)$ be the ε -free potential of the profile. At frozen (c, θ) and $\kappa = 0$ the orbit through amplitude W is, in the (ξ, u) plane, the level set $\{\frac{1}{2}u^2 = \Omega - \Phi_{c,\theta}^0(\xi)\}$ at*

$$\Omega = W/\varepsilon, \quad (3.6)$$

which does not involve ε ; and $d\tau = \sqrt{\varepsilon} d\xi/u$, so every time measured along the orbit is $\sqrt{\varepsilon}$ times a quantity depending on (Ω, c, θ) alone. In particular the period and the time below the section $z = 0$ are

$$P = \sqrt{\varepsilon} P_0(\Omega, c, \theta), \quad P_{\text{dip}} = \sqrt{\varepsilon} D_0(\Omega, c, \theta). \quad (3.7)$$

Proof. Substituting $z = \varepsilon\xi$ in Eq. (2.4) gives $m_\theta^\varepsilon(z) = m_\theta(\xi)$ and $M_\theta^\varepsilon(z) = \varepsilon M_\theta(\xi)$, so $\Phi_{c,\theta}(z) = \varepsilon \Phi_{c,\theta}^0(\xi)$. Then $W - \Phi_{c,\theta}(z) = \varepsilon(\Omega - \Phi_{c,\theta}^0(\xi))$ and $\frac{1}{2}v^2 = \varepsilon \cdot \frac{1}{2}u^2$, which is the stated level set. Equation (2.1) gives $d\tau = dz/v = \varepsilon d\xi/(\sqrt{\varepsilon}u) = \sqrt{\varepsilon} d\xi/u$, so every elapsed time is $\sqrt{\varepsilon}$ times an integral over ξ of a function of (Ω, c, θ) . Taking it over the whole orbit and over its part with $\xi \leq 0$ gives Eq. (3.7). $\square$

So ε still sets no shape, only a clock — but the clock is read at amplitude $\Omega = W/\varepsilon$, which is what the scaling of Eq. (2.4) achieves. An oscillation of amplitude of order one is read at $\Omega \rightarrow \infty$, where the profile potential is dominated by its linear term and the orbit is a long ballistic arc with a brief rebound: P_0 grows like $2\sqrt{2\Omega}$ while D_0 grows only logarithmically, so by Eq. (3.7)

$$P \rightarrow 2\sqrt{2W}, \quad \frac{P_{\text{dip}}}{P} = \frac{D_0(W/\varepsilon)}{P_0(W/\varepsilon)} \rightarrow 0, \quad (3.8)$$

the period tending to a limit of order one and the dip fraction to zero. For the exponential profile $m_\theta = e^{-\xi}$ both are explicit, $D_0/P_0 = \log(4\Omega/c)/(2\Omega)(1 + o(1))$, so that

$$\frac{P_{\text{dip}}}{P} = \frac{\varepsilon}{2W} \log \frac{4W}{c\varepsilon} (1 + o(1)). \quad (3.9)$$

Had z been measured in layer thicknesses, Ω would have been W itself and D_0/P_0 a fixed number at fixed amplitude.

Lemma 3.4 (The impulse of a pass is its entry speed). *Let a dip occupy $[\alpha, \beta]$ with $z(\alpha) = z(\beta) = 0$, and write $I = \int_\alpha^\beta m_\theta^\varepsilon(z) d\tau$ for its impulse and $P_{\text{dip}} = \beta - \alpha$. Then, exactly,*

$$cI = (v(\beta) - v(\alpha)) + P_{\text{dip}} + \int_\alpha^\beta \kappa v d\tau. \quad (3.10)$$

At frozen (c, θ) the dip is elastic — $\Phi_{c,\theta}(0) = 0$, so Eq. (2.6) gives $\frac{1}{2}v^2 = W$ at both crossings and $v(\beta) = |v(\alpha)| = \sqrt{2W}$ — and then

$$\left| I - \frac{2\sqrt{2W}}{c} \right| \leq \frac{(1 + \kappa_{\max} v_{\max}) P_{\text{dip}}}{c_{\min}}. \quad (3.11)$$

Proof. Integrate Eq. (2.2) over $[\alpha, \beta]$ and rearrange: that is Eq. (3.10), with no approximation of any kind. For Eq. (3.11), substitute $v(\beta) - v(\alpha) = 2\sqrt{2W}$ and bound the last two terms of Eq. (3.10) by $(1 + \kappa_{\max} v_{\max}) P_{\text{dip}}$, then divide by $c \geq c_{\min}$. $\square$

Together Eqs. (3.8) and (3.11) are concentration stated with constants: the pass occupies a vanishing fraction of its oscillation and nonetheless delivers an impulse of order one, fixed by the speed at which the oscillation entered — the duration goes to zero and the effect does not. In the alternative scaling the same calculation returns $I = O(\sqrt{\varepsilon})$ and a pass leaves no trace.

3.3 The excursion decomposition, and the one threshold

The identities so far are either exact along the trajectory (Lemmas 3.1 and 3.4) or statements about the frozen, undamped system (Lemmas 3.2, 3.3 and 2.8). The proof must not confuse the two, so we fix the decomposition of $[0, T]$ by the trajectory's own crossings of the section and nothing else.

Definition 3.5 (Passes, arcs, and the fast part). Fix $W_0 \in (0, W_{\max}]$ and put $v_0 = \sqrt{2W_0}$. A *pass* is a maximal subinterval of $[0, T]$ on which $z \leq 0$; an *arc* is a maximal subinterval on which $z \geq 0$. A pass or arc is called *large* — a *bubble* or a *ballistic arc* respectively — if the amplitude at the time it *begins* is at least W_0 . The *fast part* is the union of all the others; it is defined as a complement, and not as the set where the amplitude is below W_0 ,

because W may rise within an excursion when the control pumps it. What is true is that the amplitude on a fast excursion begins below W_0 and can have drifted by no more than $\Delta := (A + BW_{\max})P_1$ by Eq. (3.1), where P_1 bounds the duration of one fast excursion; hence

$$|v| \leq v_1 := \sqrt{2(W_0 + \Delta + |\Phi_{\min}|)}, \quad \Phi_{\min} = \min_z \Phi_{c,\theta}(z) = \varepsilon \Phi_{c,\theta}^0(\xi_*) = O(\varepsilon), \quad (3.12)$$

by Eq. (2.6) and Lemma 2.8, and $\Delta \rightarrow 0$ with W_0 because P_1 does. Nothing here refers to a periodic orbit: a pass is an interval between two crossings of a section, and that is all.

What the frozen system supplies is a *comparison*, used in one place only — to bound how long a pass can last, which is Lemma 3.6. Everything else descends from Eqs. (3.10), (3.12) and (5.1), all three of which hold along the trajectory as it is.

Lemma 3.6 (A pass is no longer than the frozen pass beneath it). *Let a pass occupy $[\alpha, \beta]$ with amplitude $W = W(\alpha) \geq W_0$ and control $c = c(\alpha)$, and set $\delta = (\kappa_{\max} v_{\max}^2 + M_* \dot{c}_{\max}) P_{\text{dip}}$, which bounds the drift of the amplitude across the pass by Eq. (3.1). Then*

$$P_{\text{dip}} \leq \sqrt{\varepsilon} \sup \left\{ D_0(\Omega', c', \theta) : \Omega' \geq \frac{W-\delta}{\varepsilon}, |c' - c| \leq \dot{c}_{\max} P_{\text{dip}} \right\}. \quad (3.13)$$

Proof. On $[\alpha, \beta]$, integrating Eq. (3.1) gives $W(\tau) \geq W - \delta$, and Assumption 2.5 gives $|c(\tau) - c| \leq \dot{c}_{\max} P_{\text{dip}}$. By Eq. (2.6), $\frac{1}{2}v(\tau)^2 = W(\tau) - \Phi_{c(\tau),\theta}(z(\tau))$, so at each altitude the speed is at least that of the frozen system at the same altitude with amplitude $W - \delta$ and the least favourable admissible c' . Since $P_{\text{dip}} = 2 \int_{z_-}^0 dz/|v|$ and the integrand decreases as $|v|$ grows, substituting that lower bound can only increase the integral, giving Eq. (3.13) through Eq. (3.7). The bound is implicit in P_{dip} through δ and closes by continuity: δ is of order P_{dip} , so δ/ε is of order $\log(1/\varepsilon)$ against W/ε of order $1/\varepsilon$, and the shift in Ω' is relatively negligible. $\square$

Equation (3.13) is the only inequality here that compares the trajectory with a frozen orbit, and it is affordable because a pass is short: over an interval of length P_{dip} the amplitude and the control cannot move by more than $O(P_{\text{dip}})$, so the comparison is local and self-improving and no adiabatic theorem over the horizon is needed. A construction asking the trajectory to track a closed orbit over many periods would need one.

With Definition 3.5 fixed, one number sets both errors. Let

$$\vartheta_1 = \sqrt{\varepsilon} \sup \{ D_0(\Omega, c, \theta) : \Omega \geq \frac{W_0}{2\varepsilon}, c \in [c_{\min}, c_{\max}], \theta \in \Theta \}, \quad (3.14)$$

the longest pass any bubble can have, the factor 2 absorbing the shift of Eq. (3.13) for all small ε . There is no second threshold: the error of averaging the fast part is set by v_1 of Eq. (3.12) directly, as Lemma 4.6 shows, so W_0 alone governs both.

Lemma 3.7 (How small the threshold makes each error). *Under Assumptions 2.1 and 2.6 the supremum in Eq. (3.14) is finite, and as $\varepsilon \rightarrow 0$ at fixed W_0 ,*

$$\vartheta_1 \rightarrow 0, \quad v_1 \rightarrow \sqrt{2W_0}.$$

For the exponential profile the supremum is attained at the left endpoint once ε is small, and

$$\vartheta_1 = \frac{2\varepsilon}{\sqrt{W_0}} \log \frac{2W_0}{c\varepsilon} (1 + o(1)).$$

Proof. Finiteness is Eq. (3.7) with Ω confined by Eq. (3.6) and Assumption 2.6. By Eq. (3.8) and $P_0(\Omega) \rightarrow \infty$ we have $\sqrt{\varepsilon} D_0(\Omega) \rightarrow 0$ uniformly for $\Omega \geq W_0/(2\varepsilon)$, which gives the first limit; for the exponential profile D_0 is eventually decreasing in Ω by Eq. (3.9), so the supremum is its value at $\Omega = W_0/(2\varepsilon)$, which is $\sqrt{\varepsilon} \cdot 2 \log(4\Omega/c)/\sqrt{2\Omega}$ evaluated there. The second limit is Eq. (3.12) with $\Phi_{\min} = O(\varepsilon)$. No monotonicity of D_0 in Ω is assumed, which is why Eq. (3.14) is a supremum. $\square$

Remark 3.8 (What the threshold trades). The two errors move oppositely in W_0 , and choosing between them is Remark 4.4. Raising W_0 admits fewer bubbles and shortens each pass, so ϑ_1 falls; but it leaves more of the motion to be averaged at a larger speed excursion, so v_1 rises like $\sqrt{2W_0}$. Since ϑ_1 carries a factor $\varepsilon \log(1/\varepsilon)$ and v_1 does not, the best threshold shrinks with ε — subject to Eq. (3.18), which Remark 4.4 shows is what really pins it. The count of bubbles moves with it: by Eq. (3.11) each has impulse at least $2\sqrt{2W_0}/c_{\max}$ less a correction $O(P_{\text{dip}})$, hence at least $\sqrt{2W_0}/c_{\max}$ once ε is small enough for that correction to be half the leading term, so Assumption 2.4 admits at most

$$N_* = \frac{c_{\max} M}{\sqrt{2W_0}} \quad (3.15)$$

of them — a bound free of ε at fixed W_0 , obtained from an exact integral identity and a budget and from nothing else, and which grows as W_0 is lowered to buy a smaller residual. This is the profile count the classical argument obtains only by contradiction and without a constant.

3.4 The bubble

Definition 3.9 (Pass map). For a bubble, let (v^-, s^-) be its data where it last crosses $z = 0$ downward and (v^+, s^+) where it next crosses upward, computed from Eqs. (2.1) to (2.2) together with the *inner* slow equation $\dot{s} = m_\theta^\varepsilon(z) \Gamma(s, c, \theta)$ — that is, Eq. (2.3) with the drift F dropped, whose contribution over a dip is at most $F_{\max} P_{\text{dip}}$. The *pass map* is

$$J_\theta(v^-, s^-) = \{ (v^+, s^+) : \text{measurable } c(\cdot) \text{ on the dip} \},$$

at the trajectory's own θ ; we drop the subscript where it is fixed.

Remark 3.10 (The reset carries ε , and what its limit is). Definition 3.9 is a finite- ε object: it is built from Eqs. (2.1) to (2.2), which carry ε inside the density, so strictly $J = J^\varepsilon$. Its ε -free limit is the one Lemma 3.4 exhibits. In the impulse variable the inner slow equation is $ds/d\mu = \Gamma(s, c, \theta)$, in which ε does not appear, and Eq. (3.10) fixes the *weighted* integral $\int c d\mu$ as $2|v^-|$ up to $O(P_{\text{dip}})$; so

$$J^0(v^-, s^-) = \left\{ \left(|v^-|, s^- + \int_0^{\mu_*} \Gamma(s(\mu), c(\mu), \theta) d\mu \right) : \begin{array}{l} c(\mu) \in [c_{\min}, c_{\max}], \\ \int_0^{\mu_*} c(\mu) d\mu = 2|v^-| \end{array} \right\}$$

involves no small parameter. Note that the horizon μ_* is *not* $2|v^-|/c$: the control may vary across the pass, so what Eq. (3.10) fixes is the weighted integral $\int c d\mu$ and not the horizon itself, and only a c constant on the pass makes the two the same. Lemma 3.11(b) with Eq. (3.11) gives $d_H(J^\varepsilon, J^0) = O(P_{\text{dip}}) = O(\vartheta_1)$. Theorem 4.2 compares against J^ε throughout, so nothing in Eq. (4.4) pays for this. That makes it a *uniform enclosure* statement, its reduced hybrid object still carrying the finite- ε reset, rather than a singular-limit theorem in the strongest sense. The singular-limit statement — $J^\varepsilon \rightarrow J^0$ in a set-valued topology, with convergence of the whole hybrid reachable family — is objectives 2 and 4, and is where we expect the deeper mathematics to sit.

Lemma 3.11 (The bubble is nearly elastic, and bounded in closed form). *Let $v^- \leq -v_0 < 0$ with $v_0 = \sqrt{2W_0}$, which every bubble satisfies because $\frac{1}{2}(v^-)^2 = W \geq W_0$. Along every orbit of Definition 3.9:*

(a) *at frozen (c, θ) the dip is elastic: since $\Phi_{c,\theta}(0) = 0$, Eq. (2.6) gives $\frac{1}{2}(v^\pm)^2 = W$ at both crossings, so $v^+ = |v^-|$ exactly;*

(b) *in general W drifts across the dip by at most $(\kappa_{\max} v_{\max}^2 + M_* \dot{c}_{\max}) P_{\text{dip}}$ by Eq. (3.1), so*

$$|v^+ - |v^-|| \leq \frac{(\kappa_{\max} v_{\max}^2 + M_* \dot{c}_{\max}) P_{\text{dip}}}{v_0};$$

(c) *the dip reaches no lower than the z_- with $\Phi_{c,\theta}(z_-) = W$, which exists and is unique by Lemma 2.8, and $|z_-| \rightarrow 0$ with ε ;*

(d) *its impulse I obeys Eq. (3.11), and $|s^+ - s^-| \leq \Gamma_{\max} I$.*

Proof. (a) On the dip the orbit is the level set of Eq. (2.6), so $\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)$ throughout; at $z = 0$ the potential vanishes because $M_\theta^\varepsilon(0) = 0$, giving $\frac{1}{2}(v^\pm)^2 = W$ with the same W at both crossings. (b) Away from frozen data W is no longer constant, and Eq. (3.1) bounds its rate by $\kappa_{\max} v_{\max}^2 + M_* \dot{c}_{\max}$ with M_* from Eq. (3.3); integrating over the dip and dividing by $|d(\frac{1}{2}v^2)/dv| = |v| \geq v_0$ gives the displayed bound. (c) is Lemma 2.8, and $z_- = \varepsilon \xi_-(\Omega)$ with $|\xi_-| = O(\log \Omega)$ by Lemma 3.3. (d) is Lemma 3.4 together with $s^+ - s^- = \int_{\text{dip}} m_\theta^\varepsilon \Gamma d\tau$. $\square$

Lemma 3.12 (The pass map, and how an error crosses it). *Let two entry data for the same $c(\cdot)$ and θ differ by $(\delta v, \delta s)$, with both $|v^-| \geq v_0$. Then the exits differ by $(\delta v^+, \delta s^+)$ with*

$$|\delta v^+| \leq (1 + \lambda_v P_{\text{dip}}) |\delta v| + \lambda_s P_{\text{dip}} |\delta s| + \nu P_{\text{dip}}, \quad |\delta s^+| \leq e^{\kappa_\Gamma I} |\delta s| + \frac{2\Gamma_{\max}}{c_{\min}} |\delta v|, \quad (3.16)$$

where I is the impulse of the pass and

$$\lambda_v = \frac{2\kappa_{\max} v_{\max}}{v_0}, \quad \lambda_s = \frac{\kappa_\Gamma v_{\max}^2}{v_0}, \quad \nu = \frac{2M_* \dot{c}_{\max}}{v_0}. \quad (3.17)$$

The speed is not an isometry: it is one only at frozen (c, θ) with $\kappa = 0$, where $\lambda_v = \lambda_s = \nu = 0$. The standing of the three constants is the point. The first two multiply the incoming error and are proportional to the pass's

duration, as the slow amplification is proportional to its own impulse. The third multiplies nothing: it is an additive injection, and additive because the pumping term of Eq. (3.1) carries M_θ^ε , whose derivative is the density and so unbounded in the band. No uniform Lipschitz constant for M_θ^ε exists along the pass and none is claimed; what exists is the uniform bound M_* of Eq. (3.3), which is what ν uses.

Proof. Parametrise the pass by the delivered impulse $\mu = \int m_\theta^\varepsilon d\tau \in [0, I]$, in which $ds/d\mu = \Gamma(s, c, \theta)$ — Lipschitz in s with constant κ_Γ and involving neither z nor v — so Gronwall over the horizon I gives $e^{\kappa_\Gamma I}$ on δs . The exit's dependence on δv enters through that horizon, as a finite difference rather than a derivative: the controls are only measurable and J is set-valued, so no derivative of the delivered impulse in the entry data need exist. Equation (3.10) gives cI as $v^+ - v^-$ plus terms of order P_{dip} , so for two entry data under the same $c(\cdot)$,

$$|I_1 - I_2| \leq \left(\frac{2}{c_{\min}} + C P_{\text{dip}} \right) (|\delta v| + |\delta s|),$$

with C in the data; combined with $|s^+(I_1) - s^+(I_2)| \leq \Gamma_{\max} |I_1 - I_2|$ this is the second term of Eq. (3.16), and the supremum over admissible $c(\cdot)$ turns it into the Hausdorff estimate. For the speed, do not use the frozen identity: Lemma 3.11(a) gives $v^+ = |v^-|$ only at frozen (c, θ) with $\kappa = 0$, while Definition 3.9 carries damping and a time-dependent c . Use the energy relation along the pass, $\frac{1}{2}(v^+)^2 - \frac{1}{2}(v^-)^2 = \int_\alpha^\beta (-\kappa v^2 - M_\theta^\varepsilon \dot{c}) d\tau$ from Eq. (3.1), and take the difference of the integrands for two trajectories under the same $c(\cdot)$. The damping term is Lipschitz in (v, s) with constants $2\kappa_{\max} v_{\max}$ and $\kappa_\Gamma v_{\max}^2$, giving the first two terms. The pumping term gives $\int [M_\theta^\varepsilon(z_1) - M_\theta^\varepsilon(z_2)] \dot{c} d\tau$, for which no Lipschitz estimate exists: a difference of antiderivatives is controlled by the integral of the density between the two altitudes, and that is unbounded in the band. What exists is $|M_\theta^\varepsilon(z_1) - M_\theta^\varepsilon(z_2)| \leq 2M_*$ by Eq. (3.3), uniformly in ε and regardless of how close the trajectories are — an additive injection, not an amplification. Dividing by $|v| \geq v_0$ gives Eq. (3.16) with Eq. (3.17), and suprema over admissible $c(\cdot)$ give the Hausdorff statement. $\square$

Equation (3.16) is best read as a matrix, and composing it over the passes is not a matter of telescoping: the entry carrying δv into δs is of order one, so the transfer is not the identity plus something small, and the round trip $\delta v \rightarrow \delta s \rightarrow \delta v$ must be closed explicitly.

Lemma 3.13 (Composing the passes). *Write $C_c = 2\Gamma_{\max}/c_{\min}$ for the order-one entry of Eq. (3.16). Let $u_k, y_k \geq 0$ satisfy, for $k = 0, \dots, N-1$ with $N \leq N_*$,*

$$u_{k+1} \leq (1 + \lambda_v P_k) u_k + \lambda_s P_k y_k + p_k, \quad y_{k+1} \leq e^{\kappa_\Gamma I_k} y_k + C_c u_k + q_k,$$

with $u_0 = y_0 = 0$, $\sum_k P_k = \Pi$ and $\sum_k I_k \leq M$. If

$$\lambda_v \Pi \leq \frac{1}{2} \quad \text{and} \quad 2\lambda_s C_c N_* \Pi e^{\kappa_\Gamma M} \leq \frac{1}{2}, \quad (3.18)$$

then, writing $U = \max_k u_k$ and $Y = \max_k y_k$,

$$U \leq 4 \sum_k p_k + 4\lambda_s \Pi e^{\kappa_\Gamma M} \sum_k q_k, \quad Y \leq e^{\kappa_\Gamma M} \left(\sum_k q_k + C_c N_* U \right).$$

Proof. Summing the first recursion and bounding $u_k \leq U$, $y_k \leq Y$ gives $U \leq \lambda_v \Pi U + \lambda_s \Pi Y + \sum_k p_k$, so the first half of Eq. (3.18) gives $U \leq 2(\lambda_s \Pi Y + \sum_k p_k)$. Iterating the second recursion and using $\sum_k I_k \leq M$ with $N \leq N_*$ gives $Y \leq e^{\kappa_\Gamma M} (\sum_k q_k + C_c N_* U)$, which is the second claim. Substituting it into the first,

$$U \leq 2 \sum_k p_k + 2\lambda_s \Pi e^{\kappa_\Gamma M} \left(\sum_k q_k + C_c N_* U \right),$$

and the coefficient of U on the right is the quantity the second half of Eq. (3.18) bounds by $\frac{1}{2}$; absorbing it gives the first claim. $\square$

Both conditions Eq. (3.18) are smallness conditions on $N_*^2 \vartheta_1$, since $\Pi \leq N_* \vartheta_1$ by Eq. (5.2), and $N_*^2 \vartheta_1 \rightarrow 0$ at fixed W_0 by Lemma 3.7; they are part of what $\varepsilon_0(W_0)$ asks for, and they are the price of the one order-one entry in Eq. (3.16). A weighted norm on $(\delta v, \delta s)$ closes the loop too, but costs a factor $\vartheta_1^{-1/2}$ on conversion and degrades Eq. (4.7).

4 The limit system, and the theorem

Definition 4.1 (The limit system). Fix $W_0 \in (0, W_{\max}]$ and put $v_0 = \sqrt{2W_0}$. The limit system $\mathcal{H}$ has state (z, v, s) on the arcs and (s) on the glide, and three modes.

Arc (flow). $\dot{z} = v$, $\dot{v} = -1 - \kappa v$, $\dot{s} = F(s, c, \theta)$, while $z \geq 0$: the ballistic arc, on which the air is not felt.

Bubble (jump). At $z = 0$ with $v \leq -v_0$ the state is replaced by $(0, v^+, s^+)$ with $(v^+, s^+) \in J(v^-, s^-)$, consuming no time.

Glide (averaged flow). On the fast part of Definition 3.5, where the amplitude is below W_0 : $\dot{s} = F(s, c, \theta) + \frac{1}{c} \Gamma(s, c, \theta)$, by Eq. (3.4), and z is not carried.

$\mathcal{R}_{\mathcal{H}}(t) \subseteq \mathbb{R}^{2+d}$ is the set of states reached at time t from $\mathcal{X}_0$ over all admissible c , all $\theta \in \Theta$ and all selections from J , with the convention that a state on the glide is carried as $(0, 0, s)$ — the glide mode does not propagate (z, v) , and Eq. (4.1) says what that costs. We write $\mathcal{R}_{\mathcal{H}}(E) = \bigcup_{t \in E} \mathcal{R}_{\mathcal{H}}(t)$.

That convention has a price, since $\mathcal{R}_{\varepsilon}(T)$ lives in $\mathbb{R}^{2+d}$ while the glide mode carries only s . Let

$$\Lambda = \bigcup_{c \in [c_{\min}, c_{\max}]} \bigcup_{\theta \in \Theta} \{(z, v) : \tfrac{1}{2}v^2 + \Phi_{c, \theta}(z) \leq W_0\}$$

be the fast states the fast part can occupy, the union being over the admissible data because Eq. (4.3) is one inclusion for the whole family. By Eq. (3.12),

$$\Lambda \subseteq \{|v| \leq v_1\} \times \{|z| \leq z_1\}, \quad z_1 = \sup\{|z| : \Phi_{c, \theta}(z) \leq W_0\}, \quad (4.1)$$

a set that shrinks with W_0 and with ε . A state the glide mode holds as s is to be read as the whole fibre $\Lambda \times \{s\}$ above it, and that is the lift Eq. (4.3) applies.

Note what the glide mode does *not* carry: by Eq. (3.4) the averaged slow field is independent of the amplitude, so no action coordinate is needed. The amplitude is an output of the reduced description. Note where ε still sits: $\mathcal{H}$ is ε -free in its flow and its guard, and carries ε only in its reset, to the order Remark 3.10 gives.

We measure the residual in $|(z, v, s)| = |z| + |v| + |s|$ and write $\mathbb{B}(\varrho)$ for the corresponding ball, $\oplus$ for the Minkowski sum. The constants are

$$C_f = F_{\max} + m_+ \Gamma_{\max}, \quad C_c = \frac{2\Gamma_{\max}}{c_{\min}}, \quad \kappa_0 = \kappa_F + \frac{\kappa_{\Gamma}}{c_{\min}}, \quad G = 4e^{\kappa_0 T + \kappa_{\Gamma} M}, \quad C_A = (\Gamma_{\max} + (\kappa_{\Gamma} C_f + L_c \dot{c}_{\max})T)K, \quad K = \frac{2}{c_{\min}} + \dots \quad (4.2)$$

with $m_+ = \sup_{\Theta} m_{\theta}(0)$ and M from Assumption 2.4. Every one is free of ε , which is the one discipline the proof keeps: inside the band, where the density is unbounded as $\varepsilon \rightarrow 0$, it is bounded by its *impulse* and never pointwise, and Assumption 2.4 bounds every impulse that occurs. The pointwise bound m_+ is used only above the section, where Assumption 2.1 makes it uniform.

Theorem 4.2 (Outer inclusion with an explicit residual). Assume Assumptions 2.1 to 2.6 and fix $W_0 \in (0, W_{\max}]$. There is an $\varepsilon_0 > 0$, depending only on the admissible data and W_0 , such that for every $\varepsilon \leq \varepsilon_0$ the following holds. Let ϑ_1 be the threshold Eq. (3.14), v_1 the speed bound Eq. (3.12), N_* the count Eq. (3.15), Λ the fibre Eq. (4.1), A_{ε} the impulse felt on the ballistic arcs bounded by Eq. (5.3), and E_v the total injection into the speed error defined with Eq. (4.4). Then

$$\mathcal{R}_{\varepsilon}(T) \subseteq \left[\underbrace{\mathcal{R}_{\mathcal{H}}([T - \varrho, T + \varrho])}_{\text{skeleton, bubbles and glide}} + \Lambda \times \{0\} \right] \oplus \underbrace{\mathbb{B}(\varrho)}_{\text{residual}}, \quad (4.3)$$

where the sum with $\Lambda \times \{0\}$ lifts the glide mode's slow states to the fast fibre they stand for, and

$$\varrho = G \left(\underbrace{F_{\max} N_* \vartheta_1}_{\text{the drift over the passes}} + \underbrace{C_A v_1}_{\text{averaging the fast part}} + \underbrace{C_c N_* E_v}_{\text{the speed error, routed through the coupling}} \right), \quad E_v = \nu N_* \vartheta_1 + c_{\max} A_{\varepsilon}, \quad (4.4)$$

All constants are those of Assumptions 2.1 to 2.6 through Eq. (4.2); none depends on the control, the selection, the initial point or ε . Taking the union over $\theta \in \Theta$ on both sides gives the statement for the whole admissible family.

Three remarks. The inclusion is over a short interval of times, of half-width the residual, because a pass of positive duration is compared with an instantaneous jump. The lift by Λ is not slack dressed as rigour: the averaged mode does not know where in the fast fibre the motion sits, and Λ is exactly that fibre, shrinking with W_0 alongside the residual. And each term of Eq. (4.4) is the error of one approximation, traced to one estimate below, so the theorem is modular and ϱ is a number rather than a rate; what the threshold does as $\varepsilon \rightarrow 0$ is Remark 4.4.

Remark 4.3 (What the theorem is, and is not, as a decomposition). [Theorem 4.2](#) is not a profile decomposition and does not establish one: it compares two trajectories, and profiles, their scales and their orthogonality appear nowhere in it. Two things are nevertheless true of it. It is the quantitative estimate any decomposition must reproduce — such a decomposition would have to yield [Eq. \(4.4\)](#), with [Eq. \(3.15\)](#) as its profile count — and it shows the concentration structure can be reached *without* that machinery, by an exact integral identity and a budget. Whether the machinery buys anything the elementary route does not is [objective 1](#) — the question we would put first at a blackboard, since the elementary route has already produced the enclosure.

Remark 4.4 (The threshold at its best, and the rate that follows). Optimising the threshold is not simply a matter of balancing the three terms of [Eq. \(4.4\)](#), because the choice is constrained by [Eq. \(3.18\)](#), and the constraint binds. Write t_{pump} for the part of the third term carrying the pumping injection ν . Then

$$\frac{t_{\text{pump}}}{g_2} = \frac{C_c N_* \nu \Pi}{2\lambda_s C_c N_* \Pi e^{\kappa_\Gamma M}} = \frac{M_* \dot{c}_{\max}}{\kappa_\Gamma v_{\max}^2 e^{\kappa_\Gamma M}}, \quad (4.5)$$

a constant free of both ε and W_0 — the factors C_c , N_* and Π cancel. So the second condition of [Eq. \(3.18\)](#), which [Lemma 3.13](#) needs, pins t_{pump} from *below*: $g_2 \leq \frac{1}{2}$ forces

$$\varrho \geq \varrho_\infty := \frac{G M_* \dot{c}_{\max}}{2 \kappa_\Gamma v_{\max}^2 e^{\kappa_\Gamma M}}, \quad (4.6)$$

independent of ε . Under [Assumption 2.5](#) as stated, therefore, this construction yields a *bounded* residual and not a vanishing one, and [Eq. \(4.6\)](#) is the floor. It is linear in $\dot{c}_{\max}$ and in nothing else that ε touches: the obstruction is the control's slew pumping the amplitude, routed into the slow error through the one order-one entry of [Eq. \(3.16\)](#).

Under [Eq. \(2.7\)](#) the floor disappears. With c piecewise constant, $\dot{c} = 0$ except at the at most T/Δ switching times, so only the passes containing a switch are pumped and the factor N_* in t_{pump} is replaced by T/Δ , which does not grow as W_0 falls. The cancellation in [Eq. \(4.5\)](#) then fails — the ratio acquires a factor $W_0^{1/2}$ — and balancing what remains gives $W_0 \asymp (\varepsilon \log(1/\varepsilon))^{1/2}$ and

$$\varrho = O\left((\varepsilon \log(1/\varepsilon))^{1/4}\right). \quad (4.7)$$

We record [Eq. \(4.7\)](#) for the order and claim nothing about its sharpness. The honest summary is that the rate is a statement about the guidance as much as about the model, and that removing the floor without [Eq. \(2.7\)](#) — by a composition argument whose loop gain does not scale with the pumping injection — is the sharpest open question the proof leaves. One further feature of [Eq. \(4.4\)](#) is worth naming, because it is where the constant rather than the order is lost. The third term carries the factor N_* , because the order-one entry of [Eq. \(3.16\)](#) is charged once per pass; the other two carry no such factor. So that term dominates ϱ at any threshold small enough for the rate, and an argument charging that entry once per unit of *budget* rather than once per pass would improve the constant by the ratio N_* without altering the structure. That, rather than the exponent, is where we would look first.

Remark 4.5 (At a fixed ε , and what the threshold then costs). Only one step of the proof asks ε to be small — the bubble count of Step 2, through $\varepsilon_0(W_0)$ — and below that threshold [Eq. \(4.4\)](#) is a number to be evaluated at the physical ε rather than an asymptotic statement about it. What it then says is that the residual is small only if ϑ_1 and v_1 both are, and [Section 7](#) shows that for entry they are not: no threshold makes the passes brief and the averaged excursion small together. The remedy is not a new error term but a narrower description. The motion near the threshold, which neither treatment serves, may be *propagated exactly* — carrying the fast pair (z, v) and the unapproximated field [Eqs. \(2.1\) to \(2.3\)](#) — and this adds nothing to ϱ because nothing is approximated, and removes the lift Λ over those windows because (z, v) is then carried. What it costs is dimension and amplification: the reduced problem carries $2 + d$ coordinates instead of d there, and G acquires the factor $e^{\kappa_0 T_E}$ for the total duration T_E of those windows, which [Eq. \(3.5\)](#) bounds by $2c_{\max} M$ once the windows are taken to be the intervals on which $m_\theta^\varepsilon \geq 1/(2c_{\max})$. So the threshold trades sharpness against dimension, not sharpness against sharpness.

Lemma 4.6 (Averaging the fast part, with its constant). *Let s solve [Eq. \(2.3\)](#) on a subinterval of the fast part and let $\bar{s}$ solve $\dot{\bar{s}} = F(\bar{s}, c, \theta) + \frac{1}{c}\Gamma(\bar{s}, c, \theta)$ with the same initial value. Then*

$$|s - \bar{s}| \leq G C_A v_1$$

with G , C_A as in [Eq. \(4.2\)](#) and v_1 as in [Eq. \(3.12\)](#); the factor G is the amplification over the horizon, and it is the same G that [Eq. \(4.4\)](#) collects once in front of every term, so C_A carries none of it.

Proof. Put $e = s - \bar{s}$. Subtracting, $\dot{e} = [F(s) - F(\bar{s})] + m_\theta^\varepsilon[\Gamma(s) - \Gamma(\bar{s})] + (m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s})$, so $|\dot{e}| \leq (\kappa_F + \kappa_\Gamma m_\theta^\varepsilon)|e| + |(m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s})|$.

Everything turns on the primitive $R(\tau) = \int_0^\tau (m_\theta^\varepsilon - \frac{1}{c})$, and it is bounded *from the equation of motion*, with no appeal to periodicity and none to a frozen orbit. By Eq. (2.2), $c m_\theta^\varepsilon - 1 = \dot{v} + \kappa v$, so $m_\theta^\varepsilon - \frac{1}{c} = (\dot{v} + \kappa v)/c$ and, integrating by parts in the first term,

$$R(\tau) = \left[\frac{v}{c} \right]_0^\tau + \int_0^\tau \frac{v \dot{c}}{c^2} d\tau + \int_0^\tau \frac{\kappa v}{c} d\tau, \quad (4.8)$$

an identity, exact for every control and every selection. On the fast part $|v| \leq v_1$ by Eq. (3.12), so $|R| \leq v_1(2/c_{\min} + (\dot{c}_{\max}/c_{\min}^2 + \kappa_{\max}/c_{\min})T) = v_1 K$. Note what has replaced what: the zero-mean cancellation that periodicity would have supplied is here the statement that v is *small* on the fast part, which is a property of the trajectory and not of an orbit.

Integrating the inhomogeneous term by parts against R ,

$$\left| \int_0^\tau (m_\theta^\varepsilon - \frac{1}{c}) \Gamma(\bar{s}) \right| \leq |R| \Gamma_{\max} + |R| \int_0^\tau \left| \frac{d}{d\tau} \Gamma(\bar{s}, c, \theta) \right| \leq v_1 K (\Gamma_{\max} + (\kappa_\Gamma C_f + L_c \dot{c}_{\max}) T),$$

using $|\dot{\bar{s}}| \leq C_f$ and, for the c -dependence that Γ carries along the motion, $|\frac{d}{d\tau} \Gamma(\bar{s}, c, \theta)| \leq \kappa_\Gamma |\dot{\bar{s}}| + L_c |\dot{c}|$ by Assumptions 2.2 and 2.5. Gronwall's inequality in integrated form multiplies this by $\exp \int_0^T (\kappa_F + \kappa_\Gamma m_\theta^\varepsilon) \leq e^{\kappa_F T + \kappa_\Gamma M} \leq G$, by Assumption 2.4 and not by a pointwise bound on the density, which gives the claim. The estimate does not accumulate over the successive excursions of the fast part: R is bounded over the whole horizon by Eq. (4.8), not excursion by excursion. $\square$

5 Proof of Theorem 4.2

Fix a threshold $W_0 \in (0, W_{\max}]$, an $\varepsilon \leq \varepsilon_0(W_0)$ as in Step 2, and an admissible trajectory $x_\varepsilon = (z, v, s)$ on $[0, T]$ with control $c(\cdot)$, selection $\theta \in \Theta$ and $x_\varepsilon(0) \in \mathcal{X}_0$.

Step 1: the budget

By Assumption 2.4, $\frac{d}{d\tau} \Theta_d(s) = \nabla \Theta_d \cdot F + m_\theta^\varepsilon \nabla \Theta_d \cdot \Gamma \leq -c_0 m_\theta^\varepsilon$, so

$$\int_0^T m_\theta^\varepsilon(z) d\tau \leq M, \quad (5.1)$$

for every control, every selection and every ε .

Step 2: the excursions, and the three counts

By Definition 3.5 the interval $[0, T]$ is the disjoint union of passes and arcs, cut at the trajectory's own crossings of $z = 0$, each classified by the amplitude where it begins. No periodicity is used and no orbit is asserted to close. Three counts follow, and each descends from Eq. (5.1) together with one exact identity.

How many bubbles. By Eq. (3.11) a bubble entered at amplitude $W \geq W_0$ delivers impulse at least $2\sqrt{2W_0}/c_{\max}$ less the $O(P_{\text{dip}})$ of Eq. (3.11). Let $\varepsilon_0(W_0) > 0$ be small enough that this correction is at most half the leading term — which by Lemma 3.7 it is, since $P_{\text{dip}} \leq \vartheta_1 \rightarrow 0$ while W_0 is fixed. Then for $\varepsilon \leq \varepsilon_0$ each bubble delivers at least $\sqrt{2W_0}/c_{\max}$, and since the impulses of disjoint passes sum to at most M by Eq. (5.1) there are at most the N_* of Eq. (3.15). This is Eq. (3.10) and a budget, and nothing else. It is also the one step that is asymptotic rather than uniform, which is why Theorem 4.2 carries $\varepsilon \leq \varepsilon_0$.

How long the passes last in total. By Lemma 3.6 each bubble has $P_{\text{dip}} \leq \vartheta_1$, so

$$\sum_{\text{bubbles}} P_{\text{dip}} \leq N_* \vartheta_1. \quad (5.2)$$

How much air the ballistic arcs discard. On an arc $z \geq 0$, and by Lemma 3.3 the impulse of a single ballistic arc is $\sqrt{\varepsilon} \int_0^{\xi^+} m_\theta^\varepsilon(\xi) d\xi/u \leq \varepsilon 2M_\infty/\sqrt{2W_0}$. The arcs interleave the bubbles, so N_* bubbles admit at most $N_* + 1$ of them, and an endpoint of $[0, T]$ falling inside an excursion adds no more:

$$A_\varepsilon := \sum_{\text{ballistic arcs}} \int m_\theta^\varepsilon d\tau \leq \frac{2\varepsilon M_\infty (N_* + 1)}{\sqrt{2W_0}}. \quad (5.3)$$

The number of passes grows as the threshold is lowered while each pass shortens like $\varepsilon \log(1/\varepsilon)$, and Eq. (5.2) says the product is controlled; neither it nor Eq. (5.3) is a compactness argument, and neither could be obtained from one.

Step 3: the three treatments exhaust the motion

By [Definition 3.5](#) every point of $[0, T]$ lies in a bubble, a ballistic arc, or the fast part — precisely the three modes of [Definition 4.1](#). Nothing is assumed: the classification is by the sign of z and by whether the amplitude starting the excursion reaches W_0 , both properties of the trajectory. What must be quantified is how good each treatment is, and that is [Lemma 3.7](#).

Step 4: a bubble is a jump, up to the drift over its pass

Let a bubble occupy $[\alpha, \beta]$ with $z(\alpha) = z(\beta) = 0$. By [Definition 3.9](#) the exit data of the inner problem from $(v(\alpha), s(\alpha))$ differ from $(v(\beta), s(\beta))$ only through the one term dropped in [Eq. \(2.3\)](#), namely the drift F , whose integral over the pass is at most $F_{\max} P_{\text{dip}}$. Hence

$$(v(\beta), s(\beta)) \in J(v(\alpha), s(\alpha)) \oplus \mathbb{B}(F_{\max} P_{\text{dip}}), \quad (5.4)$$

and $z(\alpha) = z(\beta) = 0$ exactly, so the altitude contributes nothing. $F_{\max}$ appears and not C_f because the inner problem retains the density term of [Eq. \(2.3\)](#) — the only reason this step survives a density unbounded inside the band.

Step 5: the fast part is the averaged glide

This is [Lemma 4.6](#), giving $G C_A v_1$ — the G being the horizon amplification [Eq. \(4.4\)](#) collects in front of every term, counted here once and not again in Step 6. It is not an averaging theorem over a periodic orbit: no orbit is assumed to close, the control varies, the damping acts, and [Eq. \(4.8\)](#) replaces the periodic argument's zero-mean cancellation by the smallness of v where the motion is averaged. Averaging over *several* phases, whose frequencies may cross, is a different problem, is attempted nowhere here, and is [objective 3](#).

Step 6: assembling, and why the amplification is bounded

Construct $x_{\mathcal{H}}$ by following $\mathcal{H}$ with the same c and the same θ , jumping at each bubble with the point of J nearest to x_{ε} 's exit, flowing ballistically on each ballistic arc, and flowing with the averaged field on the fast part. Write δ for the discrepancy and track its two components separately, which is what [Eq. \(3.16\)](#) is for.

On the arcs and the fast part both motions obey fields Lipschitz with constant κ_0 , giving $e^{\kappa_0 T}$ over the horizon. Across the bubbles the composition is [Lemma 3.13](#), with $u_k = |\delta v|$ and $y_k = |\delta s|$ at the k -th entry, $u_0 = y_0 = 0$ since both motions start together, $q_k = F_{\max} P_{\text{dip}, k}$ from [Eq. \(5.4\)](#), and p_k the speed injected by the pumping term and by the air the arcs discard. The slow amplification it produces is

$$\prod_{\text{bubbles}} e^{\kappa_{\Gamma} I_k} = e^{\kappa_{\Gamma} \sum_k I_k} \leq e^{\kappa_{\Gamma} M}$$

by [Eq. \(5.1\)](#). With the factor 4 of [Lemma 3.13](#) and the arcs' $e^{\kappa_0 T}$, these are the G of [Eq. \(4.2\)](#). This is the second place the budget does work no compactness argument could: the amplification compounds over every pass, however many, and is bounded anyway, because each factor is exponential in that pass's *own* impulse and the impulses are budget-limited. The step turns on the form of [Eq. \(3.16\)](#) and not its size: a single Lipschitz constant for J would exceed 1 by a fixed amount, and its product over the passes would diverge as the threshold was lowered.

The speed error is only *injected*, never amplified by more than $1 + \lambda_v P_{\text{dip}}$, and [Lemma 3.13](#) absorbs that. It is injected twice — by the pumping term of [Eq. \(3.16\)](#), totalling $\nu \sum P_{\text{dip}} \leq \nu N_* \vartheta_1$ by [Eq. \(5.2\)](#), and on the ballistic arcs, totalling $c_{\max} A_{\varepsilon}$ by [Eq. \(5.3\)](#) — and those two are the E_v of [Eq. \(4.4\)](#). It reaches the slow error through the one entry of [Eq. \(3.16\)](#) of order one, at most once per bubble, which is the $C_c N_* E_v$ of [Eq. \(4.4\)](#) — the only term in which the number of passes appears, and the reason [Lemma 3.13](#) is needed at all.

Collecting [Eqs. \(5.2\)](#) and [\(5.4\)](#) for the bubbles, Step 5 for the fast part and [Eq. \(5.3\)](#) for the arcs,

$$|x_{\varepsilon}(T) - x_{\mathcal{H}}(\hat{T})| \leq G \left(F_{\max} \sum_{\text{bubbles}} P_{\text{dip}} + C_A v_1 + C_c N_* E_v \right),$$

which with [Eq. \(5.2\)](#) is the ϱ of [Eq. \(4.4\)](#), the comparison being made after both states are lifted to the fast fibre Λ of [Eq. \(4.1\)](#) wherever $x_{\mathcal{H}}$ is on the glide. The arrival time $\hat{T}$ differs from T because x_{ε} spends the passes where $x_{\mathcal{H}}$ spends no time, and by [Eq. \(5.2\)](#) that discrepancy is at most $N_* \vartheta_1$, which is at most $\varrho/(GF_{\max})$ and so at most ϱ whenever $GF_{\max} \geq 1$. Hence $\hat{T} \in [T - \varrho, T + \varrho]$ and [Eq. \(4.3\)](#) follows. $\square$

Remark 5.1 (Where each hypothesis was used). [Assumption 2.1](#) was used twice: in [Lemma 2.8](#), from which the shape of the well descends, and for the uniform bound m_+ on the density above the section. [Assumption 2.4](#) was used in Step 1, and everything quantitative descends from it: the pass count [Eq. \(3.15\)](#), the summed passes [Eq. \(5.2\)](#), the arc impulse [Eq. \(5.3\)](#), the time spent in appreciable air [Eq. \(3.5\)](#), the bounded amplification of Step 6,

and the two places in [Lemma 4.6](#) where a pointwise bound on the density would have failed. [Assumption 2.5](#) was used for the *a priori* amplitude bound [Eq. \(3.2\)](#) and for the $\dot{c}$ terms in [Lemma 3.6](#) and [Eqs. \(3.17\)](#) and [\(4.8\)](#) and in C_A . [Assumption 2.2](#) supplies the constants and fixes the selection along a trajectory, without which [Eq. \(3.1\)](#) would not close ([Remark 2.3](#)); [Assumption 2.6](#) supplies the rest.

Two things must be kept apart, since the proof would be wrong if they were run together. The frozen, undamped statements — [Lemmas 3.2, 3.3](#) and [2.8](#) — only *define* P_0, D_0 and the averaged field, and enter the estimates in one place, [Lemma 3.6](#), over a single short pass. Those holding along the trajectory — [Eqs. \(3.10\), \(3.12\), \(4.8\)](#) and [\(5.1\)](#) — carry everything else, and none assumes an orbit closes. There are five hypotheses and no sixth: that the three treatments cover the motion is [Definition 3.5](#), a definition.

6 The weak formulation

[Theorem 4.2](#) concerns trajectories and its proof uses no measures. The weak formulation is nevertheless where the question belongs: it renders the dynamics a *linear* constraint, which is what makes the solution set of a nonconvex family tractable at all, and it is where the loss of compactness is visible.

With $\mathcal{U} = [c_{\min}, c_{\max}]$, the *occupation measure* of a trajectory is $\mu_\varepsilon \in \mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta)$ with $\int \varphi d\mu_\varepsilon = \int_0^T \varphi(t, x_\varepsilon(t), c(t), \theta) dt$, and μ_0, μ_T are the initial and terminal measures. The topology is weak-*; the base is a compact metric space by [Assumption 2.2](#), so it is metrizable and bounded sets are relatively compact. The dynamics are then linear: for every $\varphi \in C^1$,

$$\int \varphi(T, \cdot) d\mu_T - \int \varphi(0, \cdot) d\mu_0 = \int \left(\frac{\partial \varphi}{\partial t} + \nabla \varphi \cdot (v, cm_\theta^\varepsilon - 1 - \kappa v, F + m_\theta^\varepsilon \Gamma) \right) d\mu_\varepsilon, \quad (6.1)$$

the Liouville equation of the family, whose terminal measures' supports union to the reachable set at time T . This is the occupation-measure programme of Lasserre, Henrion and collaborators [\[4, 5\]](#), and an outer bound comes from its dual: a pair (Φ, Ψ) with Φ non-decreasing along the admissible motions and at least one on $\mathcal{X}_0$, and Ψ majorising Φ at time T , encloses the reachable set in $\{\Psi \geq 1\}$. Exhibiting such a pair is a proof whether or not it is ever evaluated numerically.

[Theorem 4.2](#) hands this machinery a different problem: the object to enclose becomes $\mathcal{R}_\mathcal{H}$ — a few coordinates with a flow, one guard $z = 0$, a set-valued reset, and a jump count bounded by [Eq. \(3.15\)](#). That places it in the class of hybrid occupation-measure formulations studied in the literature [\[6\]](#), though not directly under the cited polynomial results: the reset here is set-valued, generated by an inner controlled problem with measurable controls and an integral constraint. Bringing it inside those hypotheses is part of [objective 4](#). Two things remain. [Theorem 4.2](#) compares trajectories and not measures, which is [objective 4](#); and nothing above makes $\mathcal{R}_\mathcal{H}$ tight, so what is lost in bounding it is a separate question from what is lost in the limit. Whether the limit is better taken in this topology or at the level of trajectories as in [Section 5](#) is the open question of [Section 1.1](#); the two routes are not known to be equivalent.

7 What hypersonic entry adds

The application is hypersonic entry: the reachable set of an entry vehicle, for any vehicle and any corridor, with both entering only as data. It supplies the singular perturbation, the limit system and the concentrating events in one system, and it is not already covered. The exact method solves a Hamilton–Jacobi–Isaacs equation on a grid, capped in practice near four or five states [\[7\]](#), while a rigid-body entry vehicle carries thirteen and more once ablation and the aerothermoelastic modes are coupled [\[8\]](#); set-propagation methods return rigorous enclosures at ten to twenty states [\[9\]](#), but of one instance — one vehicle, one box, one horizon — with no explicit dependence on the corridor or on the width of the aerodynamic uncertainty, which a statement uniform over a family requires.

7.1 The correspondence

Measure altitude in planetary radii above the equilibrium glide, so the density is $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon)$ with the profile read in scale heights, and let $\varepsilon = H_s/R_e$ be the thin-atmosphere ratio, whose square root is the ratio of the oscillation period about the glide to the slow time $\sqrt{R_e/g}$. Then [Eqs. \(2.1\)](#) to [\(2.2\)](#) are the vehicle's vertical motion: v is the rate of climb, c the vertical lift fraction $(L/D) \cos \sigma$ normalised so that $cm_\theta^\varepsilon = 1$ is the glide balance — lift against weight less centrifugal relief — and κ the phugoid damping, which does not scale with ε . The slow state s collects the energy, the range and the heading, F carries the gravitational terms and Γ the drag. The selection θ chooses the density profile and the aerodynamic coefficients from an enclosure, one choice per flight by [Assumption 2.2](#), while the control still ranges over a set at each instant — so [Eqs. \(2.1\)](#) to [\(2.3\)](#) are a differential inclusion in the control over a family indexed by the uncertainty [\[10\]](#), and the vehicle enters only

through dimensionless ratios, each quantified over a range for a class and never evaluated at a design. The three regimes are the content of the model rather than an analogy: the minimum of the well is the equilibrium glide, at height of order ε ; a small oscillation is the phugoid, of period $\sqrt{\varepsilon}$ and damping ratio of the same order, so it persists for the whole entry; and a large one is a skip, of period of order one, rebounding in a time of order $\varepsilon \log(1/\varepsilon)$ by [Lemma 3.3](#) and delivering an impulse of order one by [Lemma 3.4](#).

[Assumption 2.4](#) is the hypothesis that transfers exactly. The specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$, satisfies $\dot{\mathcal{E}} = -P$ with $P = DV/m \geq 0$ the drag power, because neither the lift nor the Coriolis force does work; taking $\Theta_d = \mathcal{E}$ gives $\nabla \Theta_d \cdot F = 0$ exactly and $\nabla \Theta_d \cdot \Gamma \leq -c_0$ with c_0 fixed by lower bounds on the drag coefficient and the speed. The remaining hypotheses are conditions on the corridor: bounded away from circular speed, so that the gravity term normalises; bank bounded away from ninety degrees, giving $c \geq c_{\min} > 0$; a glide altitude existing for every admissible bank, which is the last clause of [Assumption 2.1](#); and bounded away from the poles, where the chart is singular.

7.2 Why the residual is not small at the physical ε

[Theorem 4.2](#) applies to entry, but its residual is not small, and the reason is quantitative. A pass entered at flight-path angle γ_e crosses a scale height in a time of order ε/γ_e on the slow time scale and levels out under a load of order γ_e^2/ε relative to gravity, so the entry angle enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman variables [11]. For the Earth $\varepsilon = H_s/R_e \approx 1.3 \times 10^{-3}$, so that $\sqrt{\varepsilon} \approx 0.036$ radians, about two degrees — while entry angles are a few degrees. So in the language of [Eq. \(3.14\)](#), no threshold W_0 makes ϑ_1 and v_1 small together: the amplitude at which a pass becomes brief is the amplitude at which the excursion being averaged becomes large. [Remark 4.5](#) rather than [Eq. \(4.7\)](#) is what applies, and it costs dimension rather than a new error term. Note also that entry guidance is issued on a finite interval, so [Eq. \(2.7\)](#) rather than the absolutely continuous form of [Assumption 2.5](#) is the physically apt hypothesis — which by [Remark 4.4](#) is what keeps the residual vanishing at all.

8 Plan of work

[Theorem 4.2](#) is Theorem C for one model, and it is discharged. Four obligations remain, and we state each with the obstruction it has to overcome. The first and fourth are the analysis; the second and third carry the model to the application.

Objective 1. Theorem A: the profile decomposition, with a constant

We must prove the decomposition itself: that every sequence of trajectories with $\varepsilon_n \rightarrow 0$ admits, along a subsequence, a splitting into a skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder small in a stated norm. The classical route iterates the dichotomy of the concentration-compactness principle [1] and, resolving each profile at its own scale, extracts a pairwise orthogonal family [2, 12]; orthogonality is free here, the scale of a profile being the width of a pass and its centre the time the pass occurs. What we expect to add is the constant: in that theory the count above a given size is finite because a conserved quantity is, by contradiction, where [Assumption 2.4](#) bounds it *in advance* and explicitly in [Eq. \(3.15\)](#). The objective must also say what the decomposition would buy, and four candidates are worth testing. It may give the *limit object* rather than a comparison — convergence of the family itself, which [Theorem 4.2](#) does not assert. It may remove the threshold W_0 , and with it the ε_0 of [Theorem 4.2](#), both of which exist only because the classification is made by hand. It may supply the topology in which the remainder is small, which the elementary route never needs and so never produces. And it may be what carries the support-level statement of [Section 1.1](#), the one place the trajectory route stops. If none of the four survives examination, the honest conclusion is that the machinery is the right language rather than a necessary tool — itself a result about the method. Before any of that the objective must become a statement about an actual space, and it is not one yet. Four things need specifying: the ambient set of trajectory objects and the group acting on it, here time translation with the rescaling of a pass to its own width; what convergence there means; what *orthogonal* means for two profiles, which we expect to be free, profiles at distinct times having disjoint supports after rescaling; and in what sense the remainder is small. For the last we propose the share of dissipation held by any window of fixed rescaled width — the quantity in which such a remainder should be small, and in which it is not small in the conserved quantity [2, p. 136]. But that is so far a functional on trajectories, not yet shown to be a norm or to metrize the convergence the extraction produces; until it is, the analogy with the dispersive theory is a guide rather than a theorem.

Objective 2. The statement at a fixed ε , and a time-varying atmosphere

Theorem 4.2 holds for entry with a residual Section 7 shows is not small at the physical ε . We must prove what Remark 4.5 describes: the motion near the threshold carried by the unapproximated field, the reduced problem acquiring the fast pair over a duration Eq. (3.5) bounds, and the residual gaining no new term. That is a propagation over a bounded time rather than an asymptotic estimate, so what needs controlling is its growth, explicit in the interval widths. Two further obligations join it. The reset must be replaced by the ε -free J^0 of Remark 3.10, with the cost written down. And the selection, fixed along each trajectory by Assumption 2.2 for the reason Remark 2.3 gives, must be allowed to vary: the budget is indifferent, being a common Lyapunov functional, so it is the fast half of the argument that needs rebuilding. We must state a class of time-varying selections for which the amplitude law survives — bounded variation in altitude, monotonicity where it holds, which is what the physics supplies — and carry it through Lemmas 3.1, 3.6, 3.11 and 3.12. This is a genuine enlargement of the enclosure and not a sharpening of its constants.

Objective 3. The remainder: motions that do not concentrate

A profile decomposition separates what concentrates from what does not; the second part is the remainder, and the entry problem has motions of that kind which Eqs. (2.1) to (2.3) does not carry. The attitude and the elastic modes oscillate for as long as there is air and nothing about them concentrates, so they are not profiles at all. The classical answer is averaging [13], which for *several* phases reaches no bound: it gives an order, holds only away from resonance while the attitude frequency rises from zero at every pass, and the results carrying a solution through one discard a set of initial conditions an outer statement cannot. We propose a Lyapunov functional in its place — to bound what these motions *do* rather than resolve what they are. They act on the translational motion only through the aerodynamic force, so the force is what must be enclosed, and the selection can carry that enclosure at no cost in dimension. Two facts are to be established. Where there is no air there is neither torque nor aerodynamic force, so the attitude cannot act on the trajectory at all. Where there is air, the damping rate goes as density times speed, while the energy identity makes the integral of density against the *cube* of the speed exactly the energy shed; dividing by the square of the largest admissible speed turns that equality into a lower bound on the integrated damping. The objective is met by a bound on the aerodynamic force error, explicit in the dimensionless groups of the class and uniform over the corridor.

Objective 4. The measure-level companion and the certificate

We must show that for every admissible sequence $\varepsilon_n \rightarrow 0$ the occupation measures of Section 6 have a subsequence converging weak-* to a measure that satisfies the Liouville equation of the limit system, with the mass crossing each pass carried on the graph of J — the convergence being subsequential, while the estimates and the structural conclusions are uniform over the admissible data — and exhibit the dual pair (Φ, Ψ) . This is what makes Theorem 4.2 a statement about the limit of the family rather than a comparison of two trajectories, and where weak-* compactness and concentration-compactness in the form resolving each profile at its own scale [1] are expected to be needed — for the structure, the constants coming from Assumption 2.4 as in objective 1.

8.1 Sequencing

Three phases rather than four parallel tracks. The first extends Theorem 4.2 to the planar entry problem, where Assumptions 2.1 and 2.4 are verified rather than assumed; the question of which of the three terms of Eq. (4.4) is the real obstruction belongs to it, and so does objective 1, since the count Eq. (3.15) is in hand and what remains is the decomposition it would organise. The second is objective 2, in which the threshold and the selection class are the two dials on one statement. The third is objective 3, to be reached early enough that a negative result is still recoverable, with objective 4 alongside it, needing only that the limit system be settled. If the Lyapunov route does not close, the retreat is a corridor on which the frequency bands are ordered, the crossings at the edges of each pass left to the enclosure of the bubble, which integrates the coupled field and needs no separation — weakening the residual without invalidating the decomposition.

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